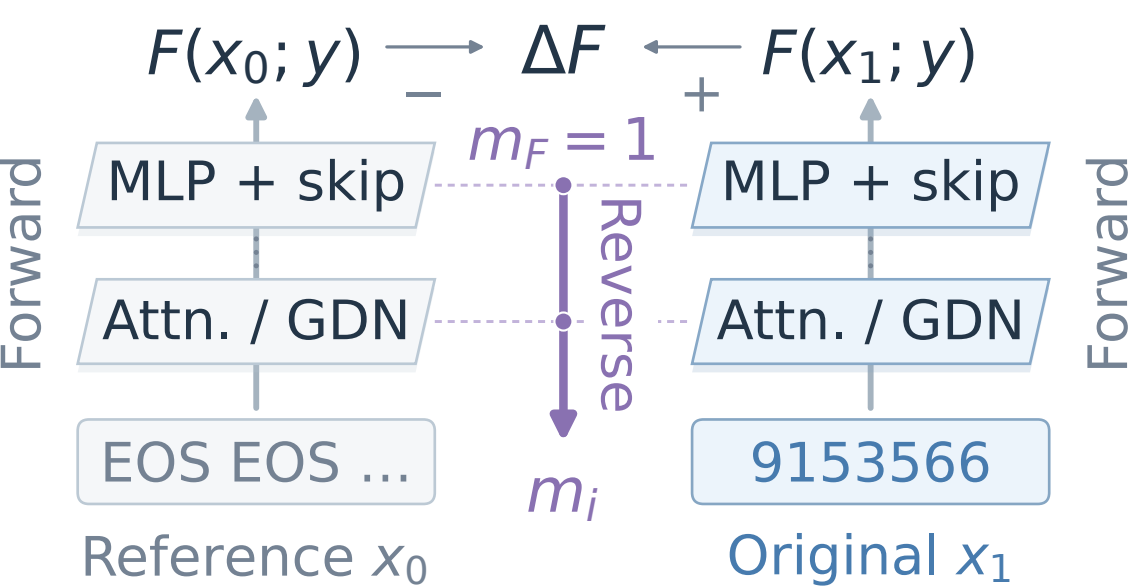


(a) From input change to credit

1 Two forward runs; fixed response y



2 Local rules

from both runs

$$\Delta h = D_f \Delta a$$

3 Reverse

propagate coefficients m

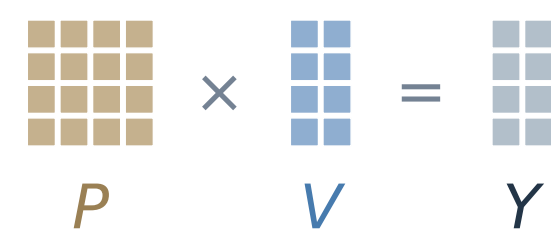
$$m_a = D_f^T m_h$$

4 Token score

$$A_i = \langle m_i, \Delta e_i \rangle$$

(b) Attention

Select and carry



0: reference; 1: original

Reverse

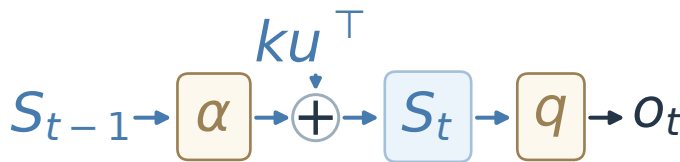
$$\Delta Y = \text{Content Selection } P_1 \Delta V + \Delta P V_0$$

Derive reverse rules

$$M_V = P_1^T M_Y \quad M_P = M_Y V_0^T$$

(c) Gated DeltaNet

Retain, write, and read



Retention: $T = \alpha S$

Reverse

$$\Delta T = \text{Content Gate } \alpha_1 \Delta S + S_0 \Delta \alpha$$

Derive reverse rules

$$M_S = \alpha_1 M_T \quad m_\alpha = \langle M_T, S_0 \rangle$$

(d) Find the names. Recover the numbers.

Qwen3.5 · retrieval example 6

CONTEXT EXCERPTS

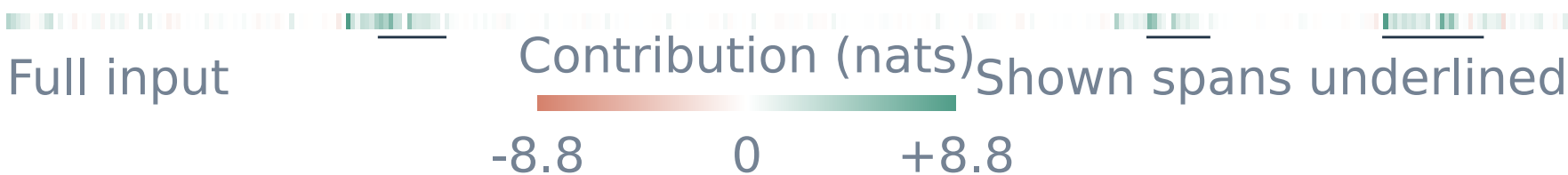
name selects · number supplies

... billowy-method is: 9937326 ·

... bright-system is: 9153566 ·

QUESTION

What are all the special magic numbers for bright-system, and billowy-method mentioned in the provided text?



Answer

9937326

billowy-method

9153566

bright-system

$$\sum_i A_i = \Delta F$$

Across all input sources